

Nonnegative and positive factorizations

Exact positive semidefinite rank of subset-intersection matrices

Difficulty: challenging **Importance:** interesting to specialist

Status: open; checked 2026-09-08

Area: structured positive semidefinite factorization

Context and notation

Write $\mathbb{S}_+^k$ for the cone of real symmetric positive semidefinite $k \times k$ matrices. For an entrywise nonnegative matrix $M \in \mathbb{R}_+^{p \times q}$, its **real positive semidefinite rank** is

$$\text{rank}_{\text{psd}}(M) = \min\{k \geq 1 : \exists A_1, \dots, A_p, B_1, \dots, B_q \in \mathbb{S}_+^k, \quad M_{ij} = \text{tr}(A_i B_j) \text{ for all } i, j\}.$$

This definition concerns a family of matrix factors indexed by the rows and columns of M .

Problem statement

For each integer $n \geq 5$, let $\mathcal{I}_n$ and $\mathcal{J}_n$ consist of the subsets of $\{1, \dots, n\}$ of cardinalities $\lfloor n/2 \rfloor$ and $\lceil n/2 \rceil$, respectively. Define the nonnegative matrix

$$M^{(n)} \in \mathbb{R}_+^{\mathcal{I}_n \times \mathcal{J}_n}, \quad M_{I,J}^{(n)} = |I \cap J|.$$

Question

Determine $\text{rank}_{\text{psd}}(M^{(n)})$ exactly as a function of n , with real symmetric factors as defined above. The family is one problem; its individual orders are not separate catalog entries.

The source supplies the bounds

$$\left\lceil \frac{\sqrt{1+8n}-1}{2} \right\rceil \leq \text{rank}_{\text{psd}}(M^{(n)}) \leq 2\lceil \sqrt{n} \rceil.$$

These matrices provide structured benchmarks for algorithms seeking small positive semidefinite factorizations.

References

Hamza Fawzi, João Gouveia, Pablo A. Parrilo, Richard Z. Robinson, and Rekha R. Thomas, *Positive semidefinite rank*, Mathematical Programming **153** (2015), 133–177, §9.1, Problem 9.2. Arnaud Vandaele, François Glineur, and Nicolas Gillis, *Algorithms for Positive Semidefinite Factorization*, Computational Optimization and Applications **71** (2018), 193–219, §4.2, preprint pp. 10–11.

Status evidence

The algorithm paper explicitly retains the unknown exact rank and uses known upper bounds in its experiments. Searches for “positive semidefinite rank Johnson scheme”, “positive semidefinite rank intersection matrix”, and “P5 psd rank” found no subsequent exact formula. The arXiv histories of both cited papers were checked; each lists only its original version. No newer explicit open-status confirmation was located.